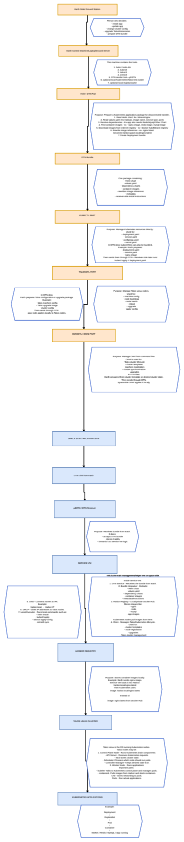
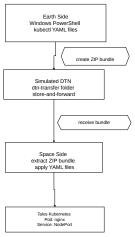
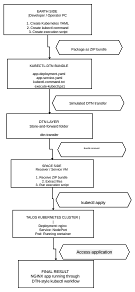

Earth Side/ Ground Station
Person who decides:
- install app
- update app
- change cluster config
- upgrade Talos/Kubernetes
- prepare DTN bundle
Earth Control Machine/Laptop/Ground Server
This machine contains the tools:
1. helm / helm-dtn
2. kubectl
3. talosctl
4. omnictl
5. DTN sender tool—µD3TN
6. optional local Kubernetes/Talos test cluster
7. optional local registry/cache
Helm- DTN Part
Purpose: Prepare a Kubernetes application package for disconnected transfer.
1. Read Helm chart. Ex- bitnami/nginx
2. Read values.yaml: Ex-replicas, image name, service type, ports
3. Resolve dependencies - Ex-app also needs Redis/Mysql/Other Chart
4. Find container images - ex - nginx image, redis image, mysql image
5. Download images from internet registry - ex- Docker hub/Bitnami registry
6. Rewrite image references - ex- nginx:latest
becomes harbor.space.local/nginx:latest
7.Create Deployment bundle
DTN Bundle
One package containing:
-Helm chart
-values.yaml
- dependency charts
- container images
- rewritten image references
- metadata
- receiver-side install instructions
KUBECTL PART
Purpose: Manage Kubernetes resources directly.
Used for:
- deployment.yaml
- service.yaml
- configmap.yaml
- secret.yaml
In DTN idea: kubectl files can also be bundled.
Example: Earth prepares:
- deployment.yaml
- service.yaml
- nginx image
Then sends them through DTN - Receiver side later runs:
kubectl apply -f deployment.yaml
TALOSCTL PART
In DTN idea:
Earth prepares Talos configuration or upgrade package.
Example:
- talos machine config
- Talos upgrade image
- cluster config
Then sends through DTN.
pace side applies locally to Talos nodes.
Purpose: Manage Talos Linux nodes.
Used for:
- machine config
- node bootstrap
- node health
- reboot
- upgrade
- apply-config
OMNICTL / OMNI PART
Purpose: Manage Omni from command line.
Omni is used for:
- Talos cluster lifecycle
- cluster templates
- machine registration
- cluster synchronization
- upgrades
In DTN idea:
Earth prepares Omni cluster template or desired cluster state.
Then sends through DTN.
Space-side Omni applies it locally.
SPACE SIDE / RECEIVER SIDE
DTN Link from Earth
µD3TN / DTN Receiver
Purpose: Receives bundle from Earth.
It does:
- accepts DTN bundle
- stores it safely
- forwards it to Service VM logic
SERVICE VM
This is the main management/helper VM on space side.
Inside Service VM:
1. DTN Service - Receives the bundle from Earth.
2. Bundle Unpacker - Extracts:
- Helm chart
- values.yaml
- dependency charts
- container images
- metadata/instructions
3. Harbor Registry - Local/private Docker Hub.
Stores images like:
- nginx
- redis
- mysql
- app images
Kubernetes nodes pull images from here.
4. Omni - Manages Talos/Kubernetes lifecycle.
Used for:
- cluster templates
- node registration
- upgrades
- Talos cluster management
5. DNS - Converts names to IPs.
Example:
harbor.local → Harbor IP
6. DHCP - Gives IP addresses to Talos nodes.
7. Local Executor - Runs local commands such as:
- helm install
- kubectl apply
- talosctl apply-config
- omnictl sync
HARBOR REGISTRY
Purpose: Stores container images locally.
Example: Earth sends nginx image.
Service VM loads it into Harbor:
harbor.local/nginx:latest
Then Kubernetes uses:
image: harbor.local/nginx:latest
Instead of:
image: nginx:latest from Docker Hub
TALOS LINUX CLUSTER
Talos Linux is the OS running Kubernetes nodes.
Talos nodes may be:
1. Control Plane Node - Runs Kubernetes brain components:
- API Server - Receives Kubernetes requests.
- etcd stores cluster state.
- Scheduler Chooses which node should run pods.
- Controller Manager- Keeps desired state true.
2. Worker Node - Runs applications.
Important parts:
- kubelet- Talks to Kubernetes control plane and manages pods.
- containerd- Pulls images from Harbor and starts containers.
CNI - Gives networking to pods.
Pods - Run actual applications.
KUBERNETES APPLICATIONS
Example:
Deployment
↓
ReplicaSet
↓
Pod
↓
Container
↓
NGINX / Redis / MySQL / App running


Earth Side
Windows PowerShell
kubectl YAML files
create ZIP bundle
Simulated DTN
dtn-transfer folder
store-and-forward
receive bundle
Space Side
extract ZIP bundle
apply YAML files
Talos Kubernetes
Pod: nginx
Service: NodePort


EARTH SIDE
|Developer / Operator PC
1. Create Kubernetes YAML
2. Create kubectl command
3. Create execution script
Package as ZIP bundle
KUBECTL-DTN BUNDLE
app-deployment.yaml
app-service.yaml
kubectl-command.txt
execute-kubectl.ps1
Simulated DTN transfer
DTN LAYER
Store-and-forward folder
dtn-transfer
Bundle received
SPACE SIDE
Receiver / Service VM
1. Receive ZIP bundle
2. Extract files
3. Run execution script
kubectl apply
TALOS KUBERNETES CLUSTER |
|
Deployment: nginx
Service: NodePort
Pod: Running container
Access application
FINAL RESULT
NGINX app running through
DTN-style kubectl workflow